3SAT $\longrightarrow$ zero-finding with constant Jacobian

Source gadget: Alpöge's announced Jacobian-conjecture counterexample · x.com/___alpoge___/status/2079028340955197566

CLAIM. Given a 3CNF formula ϕ, construct a degree-at-most-7 map K_ϕ with

$$K_\phi^{-1}(0) \neq \varnothing \iff \phi \in \mathrm{SAT}, \qquad \det JK_\phi \equiv (-3456)^{n+m}.$$

1. The gadget. Fix $H : \mathbb{R}^3 \to \mathbb{R}^3$ with $\det JH \equiv -3456$, $H^{-1}(0) = \{r_0, r_+, r_-\}$, and one omitted target $\Delta \notin \mathrm{im}(H)$. The decoder $\beta(x, y, z) = x^2$ reads r_0 as 0 and $r_\pm$ as 1.

2. One clause and the switch. Define the two-input map $T(v, s) = H(v) - (1 - s)\Delta$. The equation $T(v, s) = 0$ asks H to output $(1 - s)\Delta$: $s = 1$ asks for 0 (possible), while $s = 0$ asks for Δ (impossible). For example, if $C_j = X_a \vee \neg X_b \vee X_c$, set

$$s_j = 1 - (1 - \beta(w_a))\, \beta(w_b)\, (1 - \beta(w_c)).$$

This is 1 exactly when the clause is true, so its block satisfies

$$\exists v_j : \ T(v_j, s_j) = 0 \iff C_j \text{ is true.}$$

3. The stack. Put one $H(w_i)$ block per variable and one $T(v_j, s_j(w))$ block per clause:

$$K_\phi = \Big(H(w_1), \ldots, H(w_n), T(v_1, s_1(w)), \ldots, T(v_m, s_m(w))\Big).$$

The H blocks encode one assignment; the T blocks vanish exactly when all its clauses are true. Hence $K_\phi = 0 \iff \phi$ is satisfiable. Finally, s only translates the output of H, so $D_v T(v, s) = JH(v)$. The full Jacobian is block lower triangular with $n + m$ copies of JH on the diagonal, giving $\det JK_\phi = (-3456)^{n+m}$ and $\deg K_\phi \leq 7$.